Different Types of Storage Media

Introduction

Computer systems use a variety of storage technologies. To understand them, we classify based on two dimensions:

- **Volatility:** Does data persist without power?
- **Hierarchy:** How fast, costly, and reliable is the storage?

Main Memory (RAM and ROM)

- Acts as the system's **working storage**, where running programs and data are kept.
- **Volatile** – contents lost when power is off.
- Faster than disks but slower than CPU caches.
- Two main types: **RAM (volatile)** and **ROM (non-volatile)**.

RAM (Random Access Memory)

- Generic term for read/write main memory.
- Access time is uniform (random access).
- Always volatile.

ROM (Read-Only Memory)

- **Non-volatile** memory – data persists without power.
- Traditionally read-only; some variants can be rewritten.
- Stores firmware and boot instructions.
- Variants include **EPROM**, **EEPROM**, and **Flash**.

EEPROM (Electrically Erasable Programmable ROM)

- **Non-volatile** memory that can be erased and reprogrammed **electrically**.
- Erased and rewritten at the **byte level** (fine granularity).
- Slower than RAM but practical compared to UV-based EPROM.
- Commonly used in:
  - Microcontrollers (to store settings/configurations).
  - Embedded devices and smart cards.
  - Earlier BIOS storage.
- Has limited write cycles (typically 10^4–10^6).
- **Flash memory is a type of EEPROM** optimized for block-level erasure and higher capacity.

Types of RAM

1. SRAM (Static RAM)

- Uses **flip-flops** to store bits.
- Very fast, but expensive and consumes more power.
- Typically used for **CPU cache memory**.

2. DRAM (Dynamic RAM)

- Uses capacitors to store bits.
- Slower than SRAM, requires periodic **refreshing**.
- Cheaper and denser; used in **main memory (RAM modules)**.

Cache Memory

- A very fast, small memory located close to CPU.
- Built using **SRAM**.
- Stores frequently accessed instructions/data to reduce CPU wait time.
- Organized in levels (L1, L2, L3).

Secondary Storage

Hard Disk Drive (HDD)

- **Magnetic disk** storage.
- Non-volatile, mechanical parts (spinning platter + read/write head).
- Cheaper, larger capacity, but slower than SSDs.

Solid-State Drive (SSD)

- Uses **Flash memory** (no moving parts).
- Non-volatile, much faster than HDD.
- More expensive per GB, but durable and efficient.

Flash Memory

- A type of **EEPROM** that can be erased and reprogrammed in large **blocks**.
- Non-volatile, used in SSDs, USB drives, and memory cards.
- Slower than SRAM/DRAM but much faster than mechanical disks.

Tertiary Storage: Disks and Tapes

Magnetic Disks

- Examples: HDDs.
- Data stored magnetically on rotating platters.
- Provide large capacity but slower access.

Optical Disks

- Examples: CD, DVD, Blu-ray.
- Data read using a laser beam.
- Mostly used for media distribution and backups.

Magnetic Tape

- Sequential access storage, used for archiving.
- High capacity (TBs per tape), low cost.

Storage Hierarchy (Overview Diagram)

CPU Cache (SRAM) – Fastest, Smallest, Volatile

Main Memory (DRAM, RAM) – Fast, Volatile

SSD / Flash Memory – Non-volatile, Very Fast

HDD / Magnetic Disks – Non-volatile, Slower

Optical Disks / Magnetic Tapes – Archival, Slowest